

10(a)	<i>any eight from:</i> 1 mechanical energy converted to electrical ; 2 cell membrane depolarises ; 3 (if at least) two hairs touched (within 35 seconds); 4 action potential occurs ; 5 action potential / depolarisation, spreads over, leaf / lobe ; 6 <i>ref. to</i> hinge / midrib, cells ; 7 H⁺/ protons, pumped out of cells / pumped into cell walls ; 8 cell wall, loosens / cross-links broken ; 9 calcium pectate dissolves (in middle lamella) ; 10 Ca²⁺ (ions) enter cells ; 11 water enters by osmosis ; 12 cells, expand / become turgid ; 13 change from convex to concave ; 14 trap shuts, quickly / in <1s / in 0.3s ;	8
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Question	Answer	Marks
10(b)	<i>any seven from:</i> 1 maintain constant internal environment ; 2 despite changes in the, internal / external, environment ; 3 changes in, factor / stimulus, detected by receptor ; 4 named factor ; e.g. temperature / water potential / blood glucose concentration 5 impulses to CNS / input ; 6 <i>ref. to</i> central control / CNS, decision ; 7 impulses to effector / output ; 8 hormones to target cells 9 named effector ; e.g. muscle / gland 10 corrective action ; 11 factor returns to set point ; 12 negative feedback ; 13 AVP ; e.g. fluctuations around set point	7